

Duolingo App Review Analysis

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Abstract

In this project, I analyzed over 2,000 real user reviews from the Duolingo Android app to uncover sentiment patterns, identify user pain points, and simulate the potential impact of product improvements. By applying techniques from natural language processing (NLP), classification modeling, and business analysis, I explored how features like 'hearts', 'ads', and 'streaks' influence user sentiment and app retention. This project bridges data science with strategic product recommen...

Introduction

Duolingo, a leader in gamified language learning, has built a reputation around habit-forming engagement strategies like streaks, leaderboards, and progress rewards. While this gamification encourages frequent use, it can also introduce friction that negatively affects the user experience.

I wanted to investigate: what exactly do users enjoy, and what do they find frustrating? To answer this, I scraped and analyzed reviews from the Google Play Store to extract insights on how users feel about Duolingo's product features and how those feelings relate to churn risk.

Methodology Overview

- Data Source: ~2,200 reviews pulled using the 'google_play_scraper' library (English language, global)
- Preprocessing: Removed short/duplicate entries, applied regex cleaning, lemmatization, and stopword filtering
- Feature Tagging: Defined keyword-based feature mappings to tag mentions of "hearts", "ads", "streaks", etc.
- Sentiment Analysis: Used VADER to classify polarity scores and label sentiments (positive, neutral, negative)
- Churn Risk Modeling: Trained a logistic regression model to classify reviews as likely to indicate churn or not
- Forecasting: Built a time-series model using Prophet to predict sentiment trends over time
- Simulation: Modeled a scenario where the "hearts" feature is relaxed or removed to quantify its impact on churn

Key Insights

- 'Hearts' consistently generated frustration. Nearly 45% of reviews mentioning it were negative. Users

described it as punitive and demotivating when they made mistakes.

- In a simulated scenario where the 'hearts' system is removed, predicted churn dropped from 46% to 29%.
- Gamification features like 'streaks' and 'leaderboards' produce mixed feedback - some users find them motivating, others stressful or anxiety-inducing.
- Sentiment varies greatly by feature. A heatmap of average sentiment showed polarizing results, helping product managers prioritize which features to tweak or defend.

Recommendations

1. A/B Test a 'Heartless' Learning Mode: Create a test group that uses Duolingo without the heart system. Compare session lengths, retention rates, and sentiment metrics.
2. Customize Gamification: Offer toggle settings for streak tracking or leaderboard participation, improving inclusivity for casual learners.
3. Create Feedback Loops for Product Monitoring: Deploy sentiment dashboards internally using similar NLP tagging pipelines to track review changes post-update.

Deliverables

- Python Code: Notebooks and scripts for preprocessing, modeling, and simulation
- Streamlit Dashboard: Interactive web app for exploring feature-wise sentiment, risk trends, and forecasts
- GitHub Repository: All assets, visuals, and documentation available at: <https://github.com/shikha-punjabi/duolingo-analysis>
- Deployment: Hosted dashboard at: <https://shikha-punjabi-duolingo-analysis.streamlit.app>

Conclusion

This project demonstrates how unstructured user feedback can be transformed into actionable business strategy. Duolingo's app is beloved for its creativity and gamification, but it also shows that poorly-tuned mechanics can cause dissatisfaction and drive users away.

By simulating product changes and forecasting user sentiment, this analysis supports proactive decision-making for product teams. It also highlights my ability to extract, process, and communicate insights - translating raw data into product-level impact.